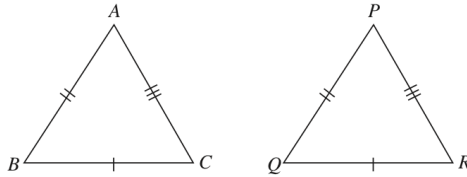
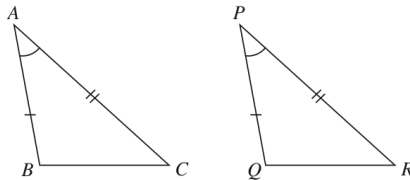


Congruent Triangles

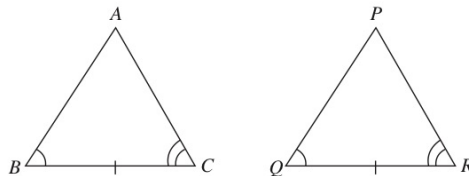
1. If $AB = PQ$, $BC = QR$ and $CA = RP$, then $\triangle ABC$ is congruent to $\triangle PQR$ (SSS Congruence Test).



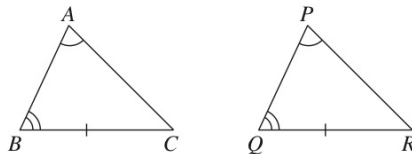
2. If $AB = PQ$, $AC = PR$ and $\hat{BAC} = \hat{QPR}$, then $\triangle ABC$ is congruent to $\triangle PQR$ (SAS Congruence Test).



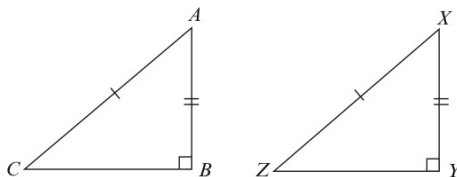
3. If $\hat{ABC} = \hat{PQR}$, $\hat{ACB} = \hat{PRQ}$ and $BC = QR$, then $\triangle ABC$ is congruent to $\triangle PQR$ (ASA Congruence Test).



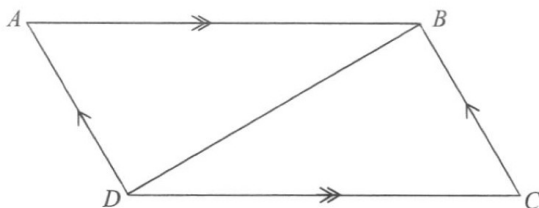
- If $\hat{BAC} = \hat{QPR}$, $\hat{ABC} = \hat{PQR}$ and $BC = QR$, then $\triangle ABC$ is congruent to $\triangle PQR$ (AAS Congruence Test).



4. If $AC = XZ$, $AB = XY$ or $BC = YZ$, and $\hat{ABC} = \hat{XYZ} = 90^\circ$, then $\triangle ABC$ is congruent to $\triangle XYZ$ (RHS Congruence Test).



In the figure below, $ABCD$ is a parallelogram with $AB \parallel DC$ and $AD \parallel BC$.



Complete the following statements for proof of congruency.

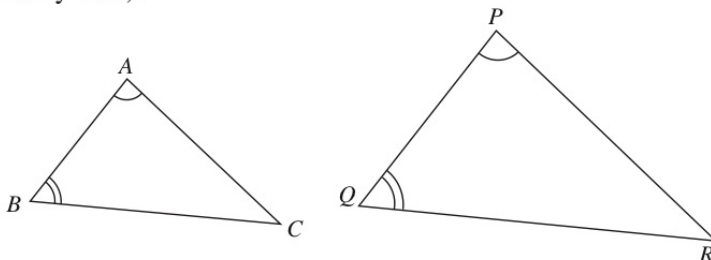
Answer

- (a) $AB = \dots\dots\dots$ (opposite sides of parallelogram are equal) [1]
 $AD = CB$ (opposite sides of parallelogram are equal)
 BD is common.
- (b) $\angle ABD = \angle CDB$ ($\dots\dots\dots$) [1]
 $\angle BAD = \angle DCB$ (opposite angles of parallelogram)
- (c) $\therefore \triangle ADB$ is congruent to $\triangle \dots\dots\dots$ [1]

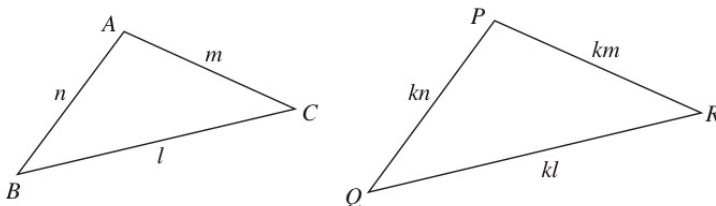
Similar Triangles

5. Two figures or objects are similar if:
- (a) the corresponding sides are proportional and,
 - (b) the corresponding angles are equal.

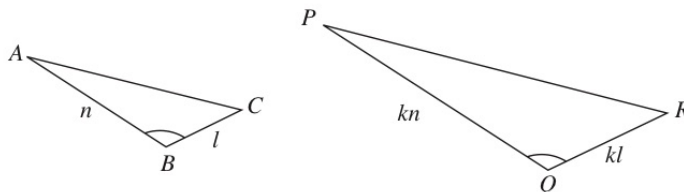
6. If $\hat{BAC} = \hat{QPR}$ and $\hat{ABC} = \hat{PQR}$, then $\triangle ABC$ is similar to $\triangle PQR$ (AA Similarity Test).



7. If $\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA}$, then $\triangle ABC$ is similar to $\triangle PQR$ (SSS Similarity Test).

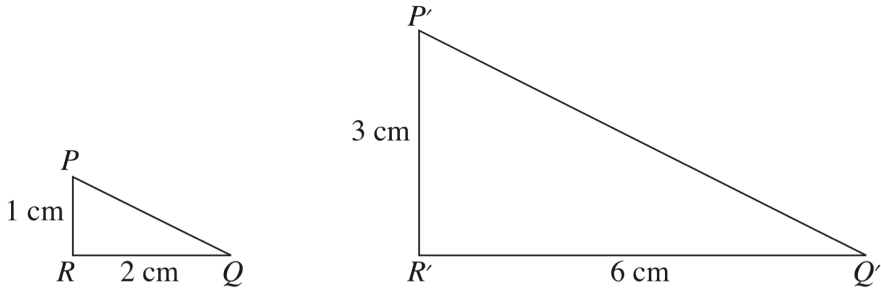


8. If $\frac{PQ}{AB} = \frac{QR}{BC}$ and $\hat{ABC} = \hat{PQR}$, then $\triangle ABC$ is similar to $\triangle PQR$ (SAS Similarity Test).



Scale Factor and Enlargement

9. Scale factor = $\frac{\text{Length of image}}{\text{Length of object}}$



$$\text{Scale factor} = \frac{P'R'}{PR} = \frac{R'Q'}{RQ} = 3$$

Similar Plane Figures

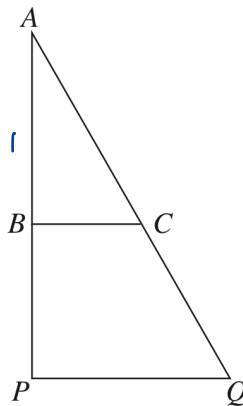
11. The ratio of the corresponding sides of two similar figures is $l_1 : l_2$.

The ratio of the area of the two figures is then $l_1^2 : l_2^2$.

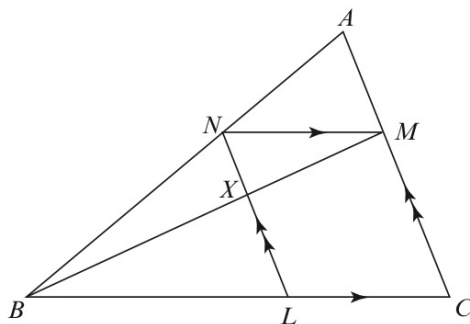
e.g. $\triangle ABC$ is similar to $\triangle APQ$.

$$\frac{AB}{AP} = \frac{AC}{AQ} = \frac{BC}{PQ}$$

$$\begin{aligned} \frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle APQ} &= \left(\frac{AB}{AP}\right)^2 \\ &= \left(\frac{AC}{AQ}\right)^2 \\ &= \left(\frac{BC}{PQ}\right)^2 \end{aligned}$$



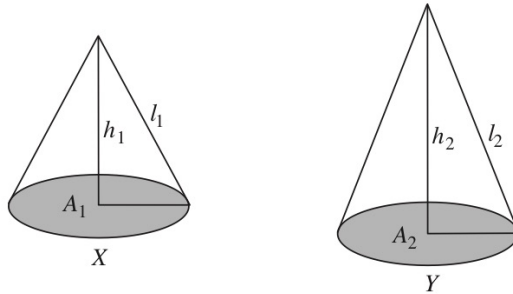
In the figure, NM is parallel to BC and LN is parallel to CA .



- (a) Prove that $\triangle ANM$ is similar to $\triangle NBL$.
- (b) Given that $\frac{AN}{NB} = \frac{2}{3}$, find the numerical value of each of the following ratios.
- $\frac{\text{Area of } \triangle ANM}{\text{Area of } \triangle NBL}$
 - $\frac{NM}{BC}$
 - $\frac{\text{Area of trapezium } BNMC}{\text{Area of } \triangle ABC}$
 - $\frac{NX}{MC}$

Triangle A and triangle B are similar. The length of one side of triangle A is $\frac{1}{4}$ the length of the corresponding side of triangle B . Find the ratio of the areas of the two triangles.

Similar Solids



- 12.** If X and Y are two similar solids, then the ratio of their lengths is equal to the ratio of their heights,

$$\text{i.e. } \frac{l_1}{l_2} = \frac{h_1}{h_2}.$$

- 13.** If X and Y are two similar solids, then the ratio of their areas is given by

$$\frac{A_1}{A_2} = \left(\frac{h_1}{h_2}\right)^2 = \left(\frac{l_1}{l_2}\right)^2.$$

- 14.** If X and Y are two similar solids, then the ratio of their volumes is given by

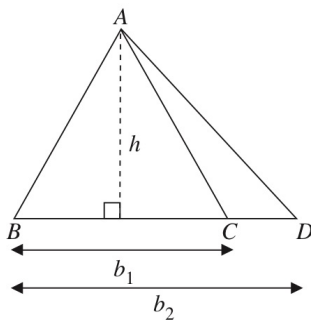
$$\frac{V_1}{V_2} = \left(\frac{h_1}{h_2}\right)^3 = \left(\frac{l_1}{l_2}\right)^3.$$

The volumes of two glass spheres are 125 cm^3 and 216 cm^3 . Find the ratio of the larger surface area to the smaller surface area.

The surface area of a small plastic cone is 90 cm^2 . The surface area of a similar, larger plastic cone is 250 cm^2 . Calculate the volume of the large cone if the volume of the small cone is 125 cm^3 .

Triangles Sharing the Same Height

16. If $\triangle ABC$ and $\triangle ABD$ share the same height h , then



$$\begin{aligned}\frac{\text{Area of } \triangle ABC}{\text{Area of } \triangle ABD} &= \frac{\frac{1}{2}b_1h}{\frac{1}{2}b_2h} \\ &= \frac{b_1}{b_2}\end{aligned}$$
